

Junction Branch Decisions and Geometry Violations

Droplet rollout model comparison

July 14, 2026

1 Purpose

This note documents the junction branch-decision diagnostic and the follow-up geometry-violation analysis. The goal is to separate two related but distinct failure modes:

- **Decision error:** the model commits to the wrong outgoing branch at the junction.
- **Physical geometry error:** the predicted droplet footprint leaves the admissible channel mask, measured by ellipse outside-channel fraction.

The analysis uses saved validation rollout predictions from:

`outputs/models/rollout_model_comparison/rollout_predictions.npz`

No model inference or retraining is involved.

2 Junction Event Definition

True junction events are identified from the true future trajectory only. A trajectory is counted as a true junction event when it has at least one valid assignment to an explicitly allowed incoming branch before committing to an outgoing branch. The outgoing branch commitment rule is:

same outgoing branch assignment for 3 consecutive evaluable rollout steps.

The outgoing branches are `left` and `right`. The incoming branch used in this validation set is `inlet`. Arbitrary non-outgoing branches are not treated as incoming, and a trajectory that begins already assigned to `left` or `right` is not counted as a true junction event.

The corrected event finder identified 4,990 true junction events:

- `inlet -> left`: 1,555 events
- `inlet -> right`: 3,435 events

3 Branch Decision Results

4 Branch Confusion Matrices

The confusion matrices below include only committed decisions: `correct` and `wrong_branch`. `no_commitment` and `unclassifiable` events are excluded from these matrices.

Table 1: Junction decision classification by model.

Model	Events	Correct	Wrong	No commit	Wrong rate	Noncommit rate
History-20	4,990	4,715	243	32	4.9%	0.6%
Markovian	4,990	4,514	387	89	7.9%	1.8%
Geometry-aware	4,990	4,498	435	57	8.8%	1.1%

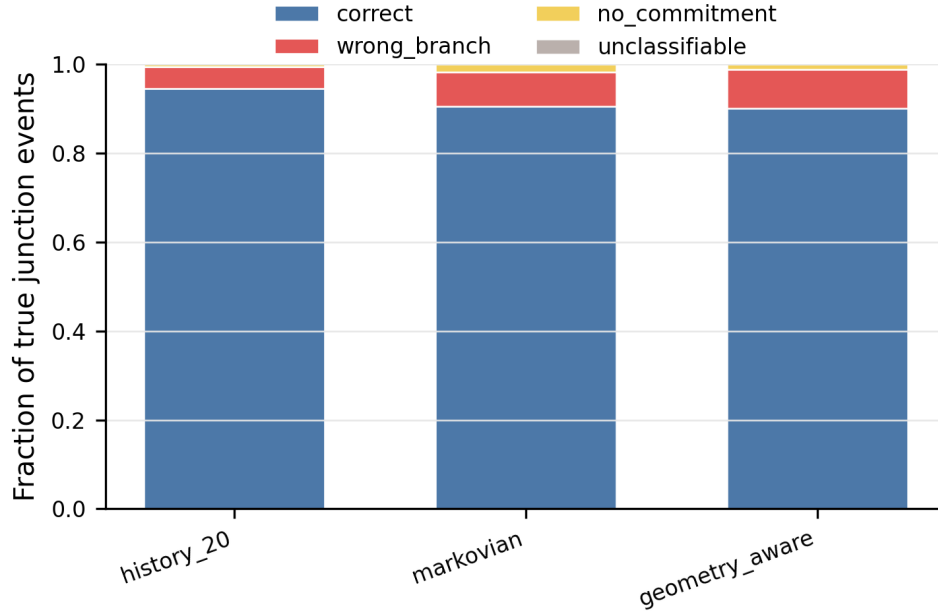


Figure 1: Decision outcome composition for true junction events. Each bar is normalized by the number of true junction events for that model.

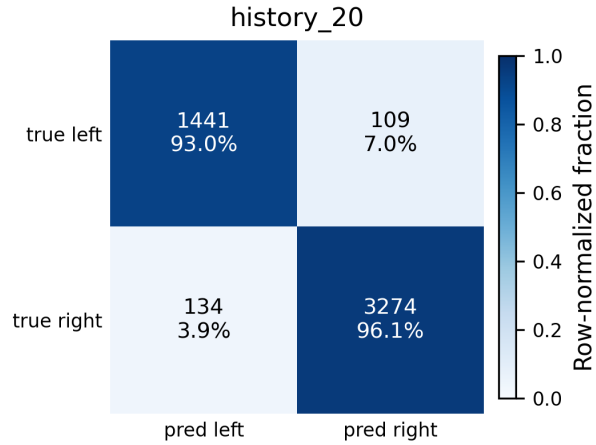


Figure 2: History-20 branch-decision confusion matrix.

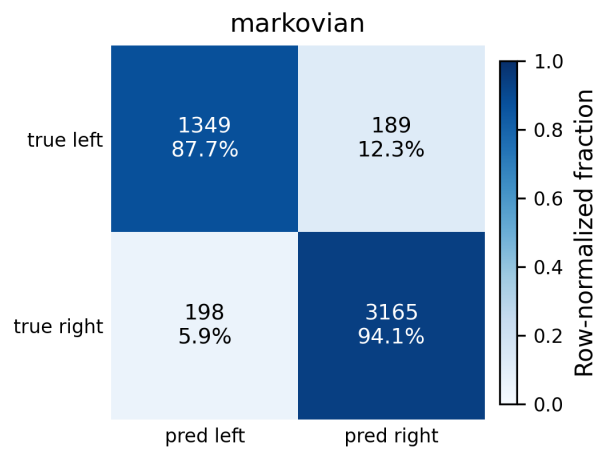


Figure 3: Markovian branch-decision confusion matrix.

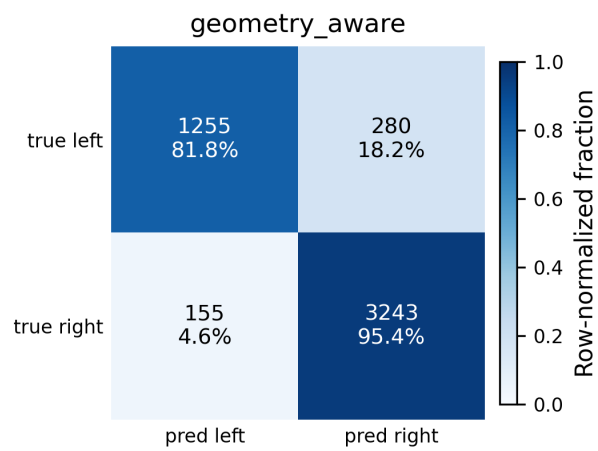


Figure 4: Geometry-aware Markovian branch-decision confusion matrix.

5 Catastrophic Channel Escape by Branch Decision

The previous generic geometry-violation summaries used a low outside-channel threshold and were not the right diagnostic for branch-decision interpretation. They are therefore not reported here. Instead, this section uses a stricter catastrophic escape criterion:

predicted ellipse outside-channel fraction ≥ 0.95 at any evaluable rollout step after predicted branch commitment.

This asks whether a committed branch prediction later becomes a complete channel escape. The analysis was run by the standalone utility:

```
python -m utils.junction_catastrophic_escape_table
```

Table 2: Catastrophic channel escape after predicted branch commitment.

Model	True branch	Pred branch	Status	N	Escape %
History-20	left	left	correct	1,441	0.35
History-20	left	right	wrong	109	1.83
History-20	right	right	correct	3,274	0.12
History-20	right	left	wrong	134	0.75
Markovian	left	left	correct	1,349	0.96
Markovian	left	right	wrong	189	9.52
Markovian	right	right	correct	3,165	0.47
Markovian	right	left	wrong	198	3.03
Geometry-aware	left	left	correct	1,255	3.35
Geometry-aware	left	right	wrong	280	0.00
Geometry-aware	right	right	correct	3,243	0.03
Geometry-aware	right	left	wrong	155	0.00

Under this stricter complete-escape definition, the geometry-aware model has no catastrophic escapes among strict wrong-branch junction events. This does not mean that the geometry-aware model never produces visually catastrophic branch tracking failures. It means that, within the strict incoming-to-outgoing junction-decision event set, its wrong-branch predictions did not reach outside-channel fraction ≥ 0.95 after predicted commitment.

6 Why Track 3050 Is Not in the Strict Junction Table

The geometry-aware video for track 3050 is a real catastrophic failure: the true droplet is on the left branch, the prediction moves to the right, and the predicted footprint eventually reaches outside-channel fraction 1.0. The specific video case is validation window 956, slot 5, track 3050.

That case is not counted in the strict junction-decision table because the true trajectory is already assigned to the left branch at rollout step 1. It does not contain an explicit `inlet -> left` transition inside the rollout window. Therefore it fails the strict true-event criterion:

explicit incoming branch assignment before outgoing branch commitment.

Track 3050 should therefore be interpreted as a **post-commitment branch tracking failure**, not as a strict incoming-to-outgoing junction decision event. This distinction matters:

- **Junction decision failures:** the droplet enters from the inlet and the model chooses the wrong outgoing branch within the rollout window.
- **Post-commitment branch tracking failures:** the droplet is already on or near a branch at the start of the rollout, and the model later diverges into another branch or leaves the channel.

7 Output Files

Primary junction-decision outputs:

```
outputs/models/rollout_model_comparison_junction_smoke/junction_decision_metrics.csv
outputs/models/rollout_model_comparison_junction_smoke/junction_decision_events.csv
outputs/models/rollout_model_comparison_junction_smoke/junction_decision_outcomes.png
outputs/models/rollout_model_comparison_junction_smoke/junction_confusion_history_20.png
outputs/models/rollout_model_comparison_junction_smoke/junction_confusion_markovian.png
outputs/models/rollout_model_comparison_junction_smoke/junction_confusion_geometry_aware.png
```

Catastrophic escape table:

```
utils/junction_catastrophic_escape_table.py
outputs/models/rollout_model_comparison/junction_decision_catastrophic_escape_table.csv
```

8 Interpretation

The decision and geometry diagnostics together show that branch choice and complete channel escape are not the same error mode. The history-based model has the best branch-decision accuracy. The Markovian model has the largest catastrophic escape rate among wrong `left -> right` decisions. The geometry-aware model shows zero catastrophic escape among strict wrong-branch junction events, but it can still fail catastrophically in post-commitment branch tracking windows such as track 3050.

The next useful diagnostic is therefore separate from the strict junction-entry analysis: explicitly measure post-commitment branch tracking failures for trajectories that begin already assigned to `left` or `right`.